

Arturo tracked the number of miles he drove each day for 15 days.

29, 34, 28, 27, 33, 36, 27, 28, 27, 25, 32, 25, 6, 31, 30

1. One value in Arturo's data set is an outlier. Which value appears to be an outlier? Explain your reasoning.

6 because the next greatest value is 25, so there is a rather large gap between the values.

2. Complete the statements.

The median of Arturo's full data set is 28.

The median of Arturo's data set with the outlier removed is 28.5.

3. Is there a significant difference between the mean of the full data set and that of the modified data set?

Mean full: ≈ 27.9

Mean modified: ≈ 29.4

The approximate difference is 1.5, so it is not significant.

4. Complete the statements.

The range of Arturo's full data set is 30. $36 - 6 = 30$

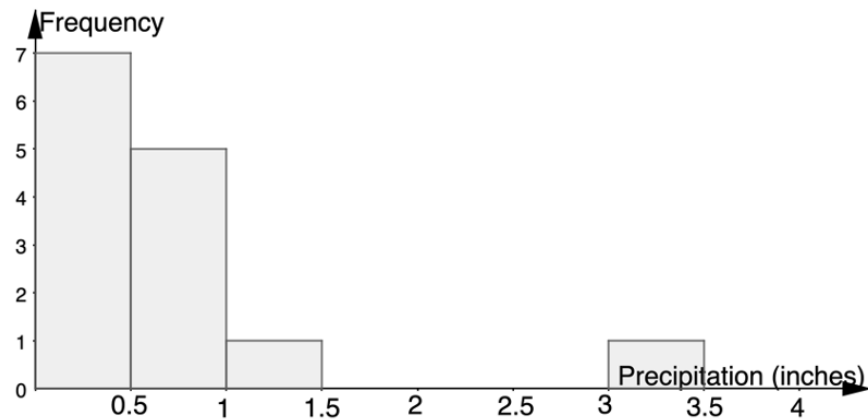
The range of Arturo's data set with the outlier removed is 11. $36 - 25 = 11$

5. Is there a significant difference between the range of the full data set and that of the modified data set?

Yes

Crystelle documented the amount of precipitation, in inches, in Bradenton, FL over the past 14 days. Her data are listed and represented in the histogram shown.

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6. How can you tell from the histogram that there is an outlier in the data set?

There is a large gap from 1.5 to 3 between the bars of the histogram.

7. What is the mean of the data set? Round to the nearest tenth of an inch.

$$\frac{8.3}{14} \approx 0.6$$

8. What is the median of the data set?

0.45

9. What impact would removing the outlier have on the mean and the median? Explain your reasoning.

The values would be the same at 0.4.

10. What impact would removing the outlier have on the IQR? Explain your reasoning.

Explanations may vary.

Removing the outlier would have less of an effect but would lower the IQR. IQR with the outlier is 0.8, and the IQR without the outlier is 0.75.